

Multivariable calculus

1. Cross and dot product

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, \quad |\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$$

$\vec{a} \times \vec{b}$ is orthogonal to both; direction by right-hand rule; magnitude = area of parallelogram spanned by $\vec{a}$ and $\vec{b}$.

2. Planes

Vector equation (normal $\vec{n}$, point $\vec{P}$): $\vec{n} \cdot (\vec{r} - \vec{P}) = 0$

Normal to level surface $F(\vec{x}) = k$ at $\vec{a}$: $\nabla F(\vec{a})$;
tangent plane: $\nabla F(\vec{a}) \cdot (\vec{r} - \vec{a}) = 0$

Plane through three points A, B, C : normal $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC}$

3. Differentiation

Gradient: $\nabla f(\vec{x}) = \langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \rangle$; perpendicular to level sets; points toward steepest ascent with magnitude $|\nabla f| = \max_{\hat{v}} D_{\hat{v}} f$.

Jacobian of $\vec{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\vec{x}$ (outputs as rows, inputs as columns):

$$J\vec{f}(\vec{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

For scalar $f : \mathbb{R}^n \rightarrow \mathbb{R}$: $Jf = (\nabla f)^T$ (a row vector).

Linear approximation to $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at $\vec{a}$ (tangent hyperplane to graph):

$$L(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

Chain rule (generic): if $\vec{r} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\vec{f} : \mathbb{R}^m \rightarrow \mathbb{R}^p$, with $\vec{x}_0 = \vec{r}(\vec{t}_0)$:

$$J(\vec{f} \circ \vec{r})(\vec{t}_0) = J\vec{f}(\vec{x}_0)J\vec{r}(\vec{t}_0)$$

In $\mathbb{R}^3$, $f \circ \vec{r} : T \subseteq \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$, with $(x, y, z) := (r_1, r_2, r_3)$:

$$\frac{d(f \circ \vec{r})}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}.$$

Directional derivative: $D_{\hat{v}} f(\vec{a}) = \nabla f(\vec{a}) \cdot \hat{v}$ (rate of change in direction $\hat{v}$).

Clairaut: $f_{xy} = f_{yx}$ when both mixed partials are continuous.

4. Curves

Arc length: $L = \int_a^b |\vec{r}'(t)| dt$; arc length function $s(t) = \int_a^t |\vec{r}'(u)| du$

Unit tangent: $\vec{T}(t) = \vec{r}'(t)/|\vec{r}'(t)|$

Curvature (rate of direction change per unit arc length):

$$\kappa(t) = \left| \frac{d\vec{T}}{ds} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}$$

Principal unit normal: $\vec{N}(t) = \vec{T}'(t)/|\vec{T}'(t)|$;
perpendicular to $\vec{T}$, points toward center of curvature.

Binormal: $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$; perpendicular to osculating plane; all three $(\vec{T}, \vec{N}, \vec{B})$ are unit vectors forming a right-handed frame.

Torsion (rate of twisting out of the osculating plane; positive = twists toward $\vec{B}$):

$$\tau(t) = -\frac{d\vec{B}}{ds} \cdot \vec{N} = \frac{(\vec{r}'(t) \times \vec{r}''(t)) \cdot \vec{r}'''(t)}{|\vec{r}'(t) \times \vec{r}''(t)|^2}$$

5. Extrema

Second derivative test ($f : \mathbb{R}^2 \rightarrow \mathbb{R}$): at a critical point, let $D = f_{xx}f_{yy} - f_{xy}^2 = \det Hf$:

$D > 0,$ $f_{xx} > 0$	$D > 0,$ $f_{xx} < 0$	$D < 0$	$D = 0$
local min	local max	saddle	inconclusive

Lagrange multipliers: at a constrained extremum of f on $\{\vec{x} : g(\vec{x}) = k\}$, the level sets of f and g are tangent, so their gradients are parallel. Solve:

$$\nabla f(\vec{x}) = \lambda \nabla g(\vec{x}), \quad g(\vec{x}) = k$$

6. Vector calculus operators

Divergence: $\operatorname{div} \vec{F} = \nabla \cdot \vec{F} = \sum_i \frac{\partial F_i}{\partial x_i}$. Measures source density at each point: positive = net outflow, negative = net inflow, zero = incompressible/solenoidal.

Curl (3D): $\operatorname{curl} \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$. Measures local rotation; direction = axis (right-hand rule), magnitude = angular speed.

Lemma: $\operatorname{div}(\operatorname{curl} \vec{F}) = 0$ for all $\vec{F}$.

2D curl: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ (the z -component of $\nabla \times \vec{F}$).

Conservative field test: on a simply connected domain, $\vec{F}$ is conservative iff equivalent conditions

- $\vec{F} = \nabla f$ for some potential f
- $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all $i \neq j$
- $\operatorname{curl} \vec{F} = 0$ in 3D (irrotational)
- $\oint_C \vec{F} \cdot d\vec{r} = 0$ for all closed C .

Finding a potential function: integrate F_1 w.r.t. x_1 to get $f + g(x_2, \dots, x_n)$; differentiate w.r.t. remaining variables and match against $F_2, F_3, \dots$ to determine g .

7. Line integrals

Scalar line integral (integrate f over arc length):

$$\int_C f \, ds = \int_a^b f(\vec{r}(t)) \left| \frac{d\vec{r}}{dt} \right| dt$$

Vector line integral (work done by $\vec{F}$ along C):

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \frac{d\vec{r}}{dt} dt$$

FTLI: if $\vec{F} = \nabla f$, then $\int_C \vec{F} \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$
– work depends only on endpoints, not the path.

8. Multiple integrals and coordinate systems

Polar (2D): $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$

$$\iint_D f \, dA = \iint f \, r \, dr \, d\theta$$

Cylindrical: same as polar + $z = z$; volume element $r \, dz \, dr \, d\theta$

$$\iiint f \, dV = \iiint f \, r \, dz \, dr \, d\theta$$

Spherical (φ = angle from z -axis, $0 \leq \varphi \leq \pi$; θ = azimuthal):

$$x = \rho \sin \varphi \cos \theta \quad y = \rho \sin \varphi \sin \theta \quad z = \rho \cos \varphi \\ \rho^2 = x^2 + y^2 + z^2$$

$$\iiint f \, dV = \iiint f \, \rho^2 \sin \varphi \, d\rho \, d\theta \, d\varphi$$

Change of variables (transformation $T_{UX} : U \rightarrow X$, i.e. $\vec{x} = T_{UX}(\vec{u})$):

$$\int \cdots \int_X f(\vec{x}) \, dx_1 \cdots dx_n \\ = \int \cdots \int_U (f \circ T_{UX})(\vec{u}) |\det JT_{UX}(\vec{u})| \, du_1 \cdots du_n$$

$|\det JT_{UX}|$ is the local area/volume stretching factor.
For polar, cylindrical, spherical the standard Jacobian determinants give the r and $\rho^2 \sin \varphi$ factors above.

9. Surfaces

Normal to parametric surface $\vec{X}(u, v)$: $\vec{N} = \vec{X}_u \times \vec{X}_v$ (orientation from right-hand rule)

For graph $z = g(x, y)$: $\vec{X}_x \times \vec{X}_y = \langle -g_x, -g_y, 1 \rangle$, so $|\vec{X}_x \times \vec{X}_y| = \sqrt{g_x^2 + g_y^2 + 1}$

Scalar surface integral (integrate f over surface area):

$$\iint_S f \, dS = \iint_D (f \circ \vec{X})(u, v) |\vec{X}_u \times \vec{X}_v| \, dA$$

Flux (net flow of $\vec{F}$ through oriented surface – sign depends on normal orientation):

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F}(\vec{X}(u, v)) \cdot (\vec{X}_u \times \vec{X}_v) \, dA$$

10. Parameterizations of common surfaces

Graph of $z = f(x, y)$:

$$\vec{X}(x, y) = \begin{bmatrix} x \\ y \\ f(x, y) \end{bmatrix} \quad \vec{X}_x \times \vec{X}_y = \begin{bmatrix} -f_x \\ -f_y \\ 1 \end{bmatrix}$$

Cylinder of radius R :

$$\vec{X}(\theta, z) = \begin{bmatrix} R \cos \theta \\ R \sin \theta \\ z \end{bmatrix}$$

Sphere of radius R :

$$\vec{X}(\varphi, \theta) = \begin{bmatrix} R \sin \varphi \cos \theta \\ R \sin \varphi \sin \theta \\ R \cos \varphi \end{bmatrix} \quad |\vec{X}_\varphi \times \vec{X}_\theta| = R^2 \sin \varphi$$

Cone $z = cr$:

$$\vec{X}(r, \theta) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \\ cr \end{bmatrix}$$

11. The big three theorems

Green's, Stokes', and the Divergence Theorem each relate an integral over a region to an integral over its boundary, reducing dimension by one.

Green's ($C = \partial D$, counterclockwise; D a planar region): circulation around the boundary = total 2D curl (rotation density) in the interior:

$$\oint_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA$$

Stokes' ($C = \partial S$, consistently oriented; S a surface): circulation around the boundary = total 3D curl through any spanning surface:

$$\oint_{\partial S} \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

S can be replaced by any other surface sharing boundary C – pick whichever is easier to integrate over.

Divergence ($S = \partial E$, outward normals; E a solid region): net outward flux through a closed surface = total divergence (source density) in the enclosed region:

$$\oiint_{\partial E} \vec{F} \cdot d\vec{S} = \iiint_E \nabla \cdot \vec{F} \, dV$$

Sign check: positive net outward flux $\leftrightarrow$ net positive sources inside.

12. Electromagnetism

$\vec{E}$ -field: It points outwards from positive charge, inward towards negative charge, with magnitude

$$\left|\vec{E}\right| = \frac{kq}{r^2} = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2}.$$

Gauss’s law: With outward-pointing (positively-oriented) $d\vec{S}$:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad \oiint_S \vec{E} \cdot d\vec{S} = \frac{q_{\text{enc}}}{\epsilon_0}$$

13. Random trig

°	rad	sin	cos	tan
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	undef

$$\cos^2 x + \sin^2 x = 1.$$